INCH-POUND

MIL-DTL-60318D (AR)
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SUPERSEDING
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DETAIL SPECIFICATION

PROPELLANT, M1
FOR USE IN CHARGE, PROPELLING, M67
FOR 105MM HOWITZER

This specification is approved for use by the US Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers M1 Propellant for one type of propelling charge designated as M67 for the 105MM Howitzer, M119A2.

1.2 Classification

The propellant shall be of the following types as specified:

Type I - Cylindrical Multi-perforated Grains
Type II - Cylindrical Single-perforated Grains

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000 or e-mailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-652	-	Propellants Solid for Cannons Requirements and Packing
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1171	-	Acceptance and Description Sheets for Propellants and Explosives
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC) DRAWINGS

9205471	Increment loading assembly
9288696	Increment loading assembly

(Copies of these drawings maybe requested online at usarmy.picatinny.ardec.list.drawing-request-help-desk@mail.mil or from US Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of documents cited in the solicitation (see 6.2).

American Society for Quality Control

ANSI/ISO/ASQC A8402-1994	-	Quality Management and Quality Assurance Vocabulary.

(Application for copies should be addressed to the American Society for Quality Control, 611 East Wisconsin Ave, Milwaukee, WI 53202.)

ASTM E178-02	-	Standard Practice for Dealing with Outlying Observations

(Address application for copies to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. The propellant shall comply with requirements in 3.2 and 3.5, and shall pass the ballistic requirements as specified in 3.3 and 3.4. In addition, the first article lot(s) shall comply with requirements for the initial production lots as specified in Table IX.

3.2 Materials. The propellant shall be M1, Type I, with basic lead carbonate for charge increments 3 through 7 and M1 Type II for charge increments 1 and 2 conforming to the requirements of MIL-STD-652 and Drawing 9288696. Alternatively, the propellant shall be M1, Type I, without lead carbonate for charge increments 3 through 7 and M1 Type II for charge increments 1 and 2 conforming to the requirements of standard MIL-STD-652 and Drawing 9205471. The components and assemblies shall comply with all requirements specified on all associated drawings and referenced specifications. (See 4.4.3.1., 6.1 and 6.15.2)

3.3 Ballistic assessment requirements. The propellant, when fired in the 105mm M119A2 howitzer, M20 cannon with 75 percent remaining life (minimum) and the M1 projectile (33.00 +/- 0.02 pounds) at 70 degrees Fahrenheit (°F) and standard weapon conditions, shall be capable of charge assessment to the zone velocities as specified in Table I. Propellant charge increment shall comply with propellant charge weight requirements as specified by Drawing (Dwg.) 9205471 or 9288696:

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TABLE I. VELOCITY AND PRESSURE

Zone*	Mean Velocity (fps) - (mps)	Mean Pressure (PSI) Minimum	Maximum
1	666.7 fps - (203.2 mps)	N/A	N/A
2	726.7 fps - (221.5 mps)	N/A	N/A
3	808.1 fps - (246.3 mps)	10,700	12,100
4	915.4 fps - (279.0 mps)	N/A	N/A
5	1057.4 fps - (322.3 mps)	N/A	N/A
6	1284.4 fps - (391.5 mps)	N/A	N/A
7	1616.8 fps - (492.8 mps)	34,100	42,700

* See paragraph 6.15.2.

3.4 Ballistic uniformity requirements. The propellant, when fired in the 105mm M119A2 howitzer, M20 cannon with 75 percent remaining life (minimum) and the M1 projectile (33.00 +/- 0.02 pounds) shall comply with the following requirements at 70°F and standard weapon conditions.

3.4.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 3.8 fps (1.16 mps) for Zone 3 or 3.4 fps (1.04 mps) for Zone 7, adjusted by the 95% upper confidence interval for the sample size tested.

3.4.2 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 48,550 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.

3.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

3.6 Propellant Aging. The minimum time from the propellant packing date to the proving ground testing shall be 90 days.

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4. VERIFICATION

TABLE II. REQUIREMENTS/VERIFICATION CROSS REFERENCE MATRIX

METHOD OF VERIFICATION		CLASSES OF VERIFICATION						
1- Analysis 2- Demonstration (end item test) 3- Examination 4- Test		A - First Article B - Conformance						
Section 3 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1	First Article	X	X	X	X	X		4.3
3.2	Materials	X		X	X	X	X	4.4.3.1
3.3	Ballistic assessment requirements		X			X	X	4.4.3.2/4.5
3.4	Ballistic uniformity requirements		X			X	X	4.4.3.2/4.5
3.5	Workmanship			X		X	X	4.4.2

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Conformance inspection (see 4.4)

4.2 Verification conditions. Unless otherwise specified, all verifications shall be performed in accordance with the test conditions specified in Section 4 of this specification.

4.3 First article.

4.3.1 Submission. Prior to initiation of sustained production, the contractor shall submit a first article sample for evaluation in accordance with the provisions of 4.3. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, except packaging, shall be obtained from the same sources as will be used in regular production. The sample shall be accompanied by certificates of analysis (see 6.2). A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted wherever there is a lapse in production for a period in excess of 90 days or wherever a change occurs in the manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspection to be performed. The sample shall be subjected by the Government to any or all of the examinations or tests specified in Section 4 of this specification and any or all requirements as specified in the applicable drawings. Test

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requirements and provisions for the initial production lots (4.5.4 and Table IX) shall be applied to the first article lot(s). In addition, the Government may also subject the first article sample to some standard safety and performance tests for propellants.

4.3.3 First article rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected.

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TABLE III. FIRST ARTICLE INSPECTION

PARAGRAPH	TITLE M1 Propellant		DRAWING NUMBER N/A NEXT HIGHER ASSEMBLY 9205471/9288696

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	Zone 7 pressure (Maximum mean pressure or maximum individual pressure)	4.4.3.2	3.3	4.5
Major				
101	Zone 3 pressure	4.4.3.2	3.3	4.5
102	Individual pressure	4.4.3.2	3.4	4.5
103	Velocity standard deviation	4.4.3.2	3.4	4.5
104	Propellant charge	Dwgs. 9205471/9288696	3.2	4.5/Balance weight
105*	Bag material	Dwgs. 9205471/9288696	3.2	AIE (see note)
106*	Bag size	Dwgs. 9205471/9288696	3.2	Visual
107*	Thread material	Dwgs. 9205471/9288696	3.2	AIE (see note)
108*	Bag marking	Dwgs. 9205471/9288696	3.2	Visual
109	Composition outside of limits specified	4.4.3.1	3.2	IAW MIL-STD-652
110	Total volatile too high	4.4.3.1	3.2	IAW MIL-STD-652
111	Moisture above/below limits	4.4.3.1	3.2	IAW MIL-STD-652
112	Compressibility too low	4.4.3.1	3.2	IAW MIL-STD-652
113	Form and dimensions	4.4.3.1	3.2	IAW MIL-STD-652
114	Heat test	4.4.3.1	3.2	IAW MIL-STD-652
115	Initial production testing	Table IX	3.1	4.5
Minor				
201	Evidence of poor workmanship	Dwgs. 9205471/9288696	3.2 and 3.5	Visual

* Failures in compliance to Major 105 to 108 will lead to rejection of test samples only, not propellant lot.
Note: Approval of inspection equipment design (AIE), see 6.10.1.

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4.4 Conformance inspection.

4.4.1 Lot formation.

The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of: (a) one or more batches of Type I, propellant or (b) one or more batches of Type II, propellant, produced by one manufacturer, in accordance with the same specification or same revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M1 propellant subjected to the same unit chemical or physical process intended to make the final product homogeneous. In addition, lot formation shall be in compliance with Paragraph 4.2.2 of MIL-STD-1916. The criteria and procedure for assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.4.2 Examination.

a. Classification of characteristics. Unless otherwise specified, sampling inspection for quality conformance characteristics listed in the classification of characteristic paragraph shall be conducted in accordance with MIL-STD-1916 using attribute sampling plans in MIL-STD-1916 (see MIL-STD-1916 for definitions of critical, major and minor classification of characteristics).

b. Alternative conformance inspection provisions. Unless otherwise specified herein or provided for in the contract, alternative conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor in accordance with Section 5 of MIL-STD-1916 when these alternate provisions provide, as a minimum, the level of verification required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.10 and 6.11). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified conformance inspection provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply.

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PARAGRAPH	TITLE	DRAWING NUMBER
4.4.2.1	Analysis of the M1 Propellant	N/A
		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Composition outside of limits specified	4.4.3.1	3.2	IAW MIL-STD-652
102	Total volatile too high	4.4.3.1	3.2	IAW MIL-STD-652
103	Moisture above/below limits	4.4.3.1	3.2	IAW MIL-STD-652
104	Compressibility too low	4.4.3.1	3.2	IAW MIL-STD-652
105	Form and dimensions	4.4.3.1	3.2	IAW MIL-STD-652
106	Heat test	4.4.3.1	3.2	IAW MIL-STD-652

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PARAGRAPH 4.4.2.2	TITLE Proving Ground Tests			DRAWING NUMBER N/A NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	Zone 7 pressure (Maximum mean pressure or maximum individual pressure)	4.4.3.2	3.3	4.5
Major				
101	Zone 3 pressure	4.4.3.2	3.3	4.5
102	Individual pressure	4.4.3.2	3.4	4.5
103	Velocity standard deviation	4.4.3.2	3.4	4.5

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PARAGRAPH	TITLE	DRAWING NUMBER
4.4.2.3	Increment Loading Assembly & Bag	9205471/9288696
		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Propellant charge	Dwgs. 9205471/9288696	3.2	4.5/Balance weight
102*	Bag material	Dwgs. 9205471/9288696	3.2	AIE (see note)
103*	Bag size	Dwgs. 9205471/9288696	3.2	Visual
104*	Thread material	Dwgs. 9205471/9288696	3.2	AIE (see note)
105*	Bag marking	Dwgs. 9205471/9288696	3.2	Visual
Minor				
201	Evidence of poor workmanship	Dwgs. 9205471/9288696	3.2 and 3.5	Visual

* Failures in compliance to Major 102 to 105 will lead to rejection of test samples only, not propellant lot.
Note: Approval of inspection equipment design (AIE), see 6.10.1.

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4.4.3 Sampling for tests

4.4.3.1 Materials. Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652. (See 3.1 and 6.7.)

4.4.3.2 Proving ground. A ballistic test quantity of 100 pounds of Type I and 100 pounds of Type II propellant shall be randomly selected from the lot for charge establishment and uniformity firings at ambient temperature. For the first three lots of propellant from any production run, a seven (7) round uniformity series shall be fired at 145°F and minus 50°F in addition to firings required to determine compliance with 3.2 at 70°F.

4.4.3.2.1 Assessment

4.4.3.2.1.1 Zone 3 Pressure. A lot of single perforated propellant shall be rejected if the average pressure at the assessed charge weight fails to meet the applicable requirement.

4.4.3.2.1.2 Zone 7 Pressure. A lot of multi-perforated propellant shall be rejected if the average pressure at the assessed charge weight fails to meet the applicable requirement.

4.4.3.2.1.3 Velocity uniformity. If the sample muzzle velocity standard deviation exceeds the requirement divided by the factor (see 6.8), at any zone, the lot shall be rejected. The factor for 7 samples and 95% confidence is 1.92.

4.4.3.2.1.4 Individual pressure. If any individual pressure observed during uniformity firings at any operating temperature (-50°F to +145°F) after correction for difference between as fired and recommended service charge weight exceeds 48,550 psi, the lot shall be rejected.

4.4.3.2.1.5 Average pressure for Multi-perforated (MP) propellant. If the average pressure at Zone 7 fails to fall within 34,100 to 42,700 psi, the lot of MP propellant shall be rejected.

4.4.3.2.1.6 Mean pressure plus standard deviation. The mean pressure at Zone 7 plus four times the standard deviation of the sample obtained during the uniformity series exceeds 48,550 psi, the lot shall be rejected.

4.5 Test Methods and Procedures

4.5.1 Proving ground tests. These tests shall be conducted at a proving ground in accordance with the applicable Acceptance Test Procedure (ATP) using inert M1 projectile (33.00 +/- 0.02 pounds) and fired from a M119A2 Howitzer with a M20 Cannon with 75 percent minimum remaining life. Prior to firing, all charges shall be temperature conditioned for a minimum of 24 hours and shall be fired within five minutes after removal from the conditioning chamber. A tolerance of - 5 °F applies to all - 50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

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Calibration charges shall be fired alternatively with the test samples and the muzzle velocity and pressure of all charges shall be corrected to standard conditions.

4.5.2 Charge weight establishment for Single-perforated (SP) propellant

4.5.2.1 Charge weight establishment for Zone 1 and Zone 2 (see Table IV). Three groups of rounds loaded with selected weights of propellant for zones 1, 2 and 2+ shall be conditioned to 70°F plus or minus 2°F for at least 24 hours and shall be fired within five minutes after removal from the conditioning chamber. The first group consists of seven (7) Zone 1 charges, the second group consists of seven (7) Zone 2 charges and the third group consists of seven (7) zone 2+ charges. From the data obtained, a correlation between velocity and charge weight shall be derived using linear regression (excluding any valid outlier), and the charge weight shall be calculated which will yield the service velocity as specified in Table I for each zones 1 and 2. This calculated charge weight shall then be loaded into rounds for zones 1 and 2 uniformity series in 4.5.2.2. Calibration rounds shall be used to correct velocity. Zone 2 calibration value shall be used to correct zone 2 and zone 2+ test round velocities. Alternatively, closed bomb firings may be substituted for ballistic firings at zones 1 and 2 for charge weight establishment only in accordance with MIL-STD-286, Method 801.1.2.

TABLE IV. CHARGE WEIGHT ESTABLISHMENT FOR ZONES 1 AND 2

No. of Rounds	Type of Round	Zone	Charge Weight
7	Calibration	2	Service
7	Test	2	Estimated SP Propellant
7	Test	2+	Estimated SP Propellant at Zone 2 w/ 105%
7	Test	1	Estimated SP Propellant
7	Calibration	1	Service

4.5.2.2 Uniformity series for Zone 1 and Zone 2 (Table V). The propellant shall be conditioned to 70°F plus or minus 2°F for at least 24 hours and shall be fired within five minutes after removal from the conditioning chamber. Firing results at 70°F shall be corrected to standard weapon/firing conditions (calibration round conditions). A seven round uniformity series shall be fired at each zone as specified in Table V. The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity versus charge weights curve established above. Combining the data obtained from both charge weight establishment and uniformity series (70°F), a correlation between velocity and charge weight shall be derived using linear regression (excluding any valid outlier), and the charge weight shall be calculated which will yield the service velocity as specified in Table I for each zones 1 and 2. The calculated charge weights shall be the recommended charge weights for zones 1 and 2.

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TABLE V. UNIFORMITY SERIES FOR SP PROPELLANT

No. of Rounds	Type of Round	Zone	Charge Weight
7	Calibration	1	Service
7	Test	1	*
7	Calibration	3	Service
7	Test	3	**
7	Test	3	***

Notes: * SP propellant weight as determined from charge weight establishment.

** Comprised of test SP propellant (charge increments 1 and 2) and MP calibration propellant (charge increments 3)

*** Comprised of test SP propellant (charge increments 1 and 2) and MP propellant (charge increment 3) to be loaded for charge increments 3-7.

4.5.2.2.1 Average pressure for SP propellant. If any lot of SP propellant fired in conjunction with MP propellant fails to meet the zone 3 pressure requirement (10,700 – 12,100 psi), the SP propellant lot shall be referred to Picatinny Arsenal: U.S. Army ARDEC, ATTN: RDAR-MEE-W, Picatinny Arsenal, NJ 07806-5000.

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4.5.3 Multi-perforated (MP) propellant assessment.

4.5.3.1 Charge weight establishment for zones 3, 4, 5, 6, and 7 (Table VI). After charge weights for charge increments 1 and 2 been established, four groups of rounds loaded with selected charge weights of propellant for charge increments 3, 5, 6, and 7 (weights selected should bracket expected service charge weight) shall be conditioned to 70°F plus or minus 2°F for at least 24 hours and shall be fired within five minutes after removal from the conditioning chamber. The first group consists of seven (7) Zone 3 charges, the second group consists of four (4) zone 5 charges, the third group consists of four (4) zone 6 charges, and the fourth group consists of seven (7) Zone 7 charges. From the data obtained, a correlation between velocity and charge weight shall be derived using linear regression (excluding any valid outlier), and the charge weight shall be calculated which will yield the service velocity as specified in Table I for each zones 3, 4, 5, 6, and 7. This calculated charge weight shall then be loaded into rounds for the uniformity series in 4.5.2.2. Calibration rounds shall be used to correct velocity.

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TABLE VI. CHARGE WEIGHT ESTABLISHMENT FOR CHARGE INCREMENTS 3-7

No. of Rounds	Type of Round	Zones	Charge Weight
7	Calibration	7	Service
7	Test	7	Estimated MP Propellant
4	Calibration	6	Service
4	Test	6	Estimated MP Propellant
4	Calibration	5	Service
4	Test	5	Estimated MP Propellant
7	Calibration	3	Service
7	Test	3	Estimated MP Propellant

4.5.3.2 Uniformity Series for zones 3-7 (Table VII). The propellant shall be conditioned to 70°F plus or minus 2°F for at least 24 hours and shall be fired within five minutes after removal from the conditioning chamber. Firing results at 70°F shall be corrected to standard weapon/firing conditions (calibration round conditions). A seven round uniformity series shall be fired at each zone as specified in Table VII. Combining the data obtained from both charge weight establishment and uniformity series (70°F), a correlation between velocity and charge weight shall be derived using linear regression (excluding any valid outlier), and the charge weight shall be calculated which will yield the service velocity as specified in Table I for each zones, 3, 4, 5, 6 and 7. The resultants shall be the recommended charge weights for charge increments 3, 4, 5, 6 and 7.

TABLE VII. UNIFORMITY SERIES FOR MP PROPELLANT

No. of Rounds	Type of Round	Zone	Charge Weight
7	Calibration	7	Service
7	Test	7	*
7	Test	5	*
7	Calibration	5	Service
7	Test	3	*
7	Calibration	3	Service

* Zones 1 and 2 final recommended charge weights. Zones 3 to 7 as determined for MP propellant from charge weight establishment.

4.5.4 Initial production. For initial production, each of the first three lots of propellant from any production runs shall be fire-tested in accordance with test conditions, sample size and performance requirements as specified in Table IX.

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Table IX. INITIAL PRODUCTION

Temp. (ºF)	Zone	Sample Size	Vel (fps) Assessed	Lot Std. Dev. of Vel. (fps)	Perm. Ind. Max Press	Mean Pressure (PSI) Min	Max
-50	7	7	N/A	N/A	48,550	N/A	N/A
70	3	7	808.1	3.8	N/A	10,700	12,100
70	7	7	1616.8	3.4	48,550	34,100	42,700
145	7	7	N/A	N/A	48,550	N/A	N/A

4.5.5 Retest. If for any reason the test agency or ARDEC considers that test conditions have detrimentally affected the test results, they shall declare the results invalid and perform a new test on additional charges as required. Prior approval shall be obtained from the technical agency before retest initiation. (See 6.4 and 6.9).

4.5.6 Referral criteria. A test lot shall be referred whenever any of the following conditions exist:

a. Calibration charges fired in any test phase fail to comply with Section 3 of this specification.
b. Calibration components are not available.
c. Number of test or calibration charges fired is less than specified.
d. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc. Each such event shall be reported in detail on applicable firing record.
e. Test lot fails velocity standard deviation or average pressure requirement at 70 degrees F and extreme value is an outlier (see 6.13).
f. For zone 3, if the absolute difference from fired charge uniformity weight and the final recommended charge weight is 8% or more (see 6.14).
g. For zone 4, if the absolute difference from fired charge uniformity weight and the final recommended charge weight is 3% or more (see 6.14).
h. For zone 5, if the absolute difference from fired charge uniformity weight and the final recommended charge weight is 3% or more (see 6.14).
i. For zone 6, if the absolute difference from fired charge uniformity weight and the final recommended charge weight is 3% or more (see 6.14).
j. For zone 7, if the absolute difference from fired charge uniformity weight and the final recommended charge weight is 3% or more (see 6.14).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense

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Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity. (See 6.5)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M1 Propellant is intended to be used for the M67 propelling charges for 105MM Howitzer. There are two alternatives:

6.1.1 Dwg. 9288695. Charge, Propelling, M67 Assembly uses M1 propellant that contains basic lead carbonate. No lead foil is used in the increment loading assembly, (Dwg. 9288696).

6.1.2 Dwg. 9205472. Charge Propelling, M67 Assembly uses M1 propellant without basic lead carbonate and is used with lead foil in the charge increment number 5 loading assembly (Dwg. 9205471).

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Acceptance and description sheets should be prepared for each lot in accordance with MIL-STD-1171.
d. Provisions for submission of first article samples.
e. Type and form of the trinitrotoluene.
f. Levels of preservation and packing.
g. Item hazard classification.
h. Requirements for submission of inspection equipment design(s).

6.3 Standard deviation. The standard deviation should be calculated with (n-1) as the devisor in a standard statistical technique.

6.4 Submission of test data. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data should be forwarded to : U.S. Army ARDEC, ATTN: AMSTA-AR-WEE-B (one copy), and to ATTN: AMSRD-AAR-QEM-C (one copy), Picatinny Arsenal, NJ 07806-5000, and AFSC - JMC: ATTN SFSJM-CDB, Rock Island, IL 61299-6000 (two copies)

6.5 Acceptable packing and marking requirements. Packing and marking requirements should be in accordance with MIL-STD-652.

6.6 Surveillance sample. Samples should be forwarded to U.S. Army ARDEC, ATTN: AMSTA-AR-WEE-B, Picatinny Arsenal, NJ 07806-5000

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6.7 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance the contractor should supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of batches of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of pounds in the lot.
g. Date submitted.
h. Eight copies of propellant description sheet containing the applicable data (see MIL-STD-1171).

The certificate should be signed by a responsible agent of the certifying organization. The initial certificate submitted should be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

6.8 Factors. The upper confidence interval factors are developed from:

$$Factor = \sqrt{\frac{n-1}{invχ^2(1-alpha, n-1)}}$$

Where, n is the sample size and $invx^2$ is the inverse chi-squared function and alpha is the chosen confidence level. This method is used to ensure that a small sample size is capable of representing that the lot meets the overall standard deviation requirement.

6.9 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000.

6.10 Approval of inspection equipment designs and equivalent test methods.

6.10.1 Submission of inspection equipment designs for Government approval. Inspection equipment designs should be submitted through the Contracting Officer to the technical agency (see 6.9) for approval. The submission should include a detailed description of the inspection equipment system, equipment operating procedure,

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calibration procedure and along with accuracy and precision data for the proposed inspection equipment system. Prior approval of the Contracting Officer is required for use of the contractor proposed inspection equipment design(s).

6.10.2 Submission of equivalent test methods for Government approval. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required

6.11 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification.

6.12 Graphite Glazing. Type I and Type II propellant shall be glazed with graphite, Grade III or IV, MIL-G-155, up to 0.010% by weight, max. Approval for the use of graphite should be referred to the procuring activity, Office of the Project Manager for Combat Ammunition Systems, ATTN: SFAE-AMO-CAS, Picatinny, NJ 07806-5000.

6.13 Outlier. The following test should be performed to determine if a single test result is an outlier. With a sample size of 7 at a significance level of 0.05, the critical value (Tc) is 0.507. Calculate a test value (Tc) as follows:

For Low Outlier: Tt = (X2-X1) / (X7-X1)

For High Outlier: Tt = (X7-X6) / (X7-X1)

Where:
X1 = test results being tested as low outlier.
X2 = test results adjacent to X1, after all the test results are ordered from smallest to largest.
X6 = test result adjacent to the largest test result.
X7 = test result being tested as high outlier.
X1 or X7 is an outlier if Tt is greater than Tc.

“6.14 Referral for charge weight deviation. If the absolute difference from fired charge uniformity weight and the final recommended charge weight is 8% or more for zone 3, 3% or more for zone 4, 3% or more for zone 5, 3% or more for zone 6, or 3% or more for zone 7, the lot should be referred to Commander, ARDEC, ATTN: RDAR-QEM-A, Picatinny Arsenal, NJ 07806-5000 for evaluation for retest and final disposition.

6.15 Subject term (key word) listing and definitions.

6.15.1 Subject term (key word) listing

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Single-perforated
Multi-perforated
Zone
Charge Increment

6.15.2 Definitions

M1	-	Type of propellant.
M67	-	Type of propelling charge.
Charge Increment	-	The single increment bag of propellant.
Zone	-	Cumulative term which refers to all the charge increments up to and including the one specified, i.e. Zone 3 includes charge increments 1, 2, and 3.

6.16 Change notations. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians
Army - AR

Preparing activity:
Army-AR
(Project 1376-2016-011)

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